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Using the Derivative to Graph Functions

Video: Using the Derivative to Graph Functions
4 min

Quiz: Using the Derivative to Graph Functions Quiz
Started

Finding Concavity with the Second Derivative

Comparing the Graphs of f(x) and f'(x)

Graphing Functions - A More Complicated Example

Using the Derivative to Graph Functions Quiz

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Due Mar 17, 11:59 PM IST

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1. What does it mean for $f'(x) > 0$ for some x ?

1 / 1 point

- ☐ The function is decreasing at x
- ☐ The function is not changing at x
- ☒ The function is increasing at x

✔ Correct
Great! When $f'(x) > 0$ then the slope of the function is positive at x

2. Where is f increasing and decreasing for $f(x) = x^3 - 3x^2 - 9x + 4$?

1 / 1 point

- ☒ f is increasing for $x < -1, x > 3$.
 f is decreasing for $-1 < x < 3$.
- ☐ f is increasing for $x < -1$.
 f is decreasing for $x > -1$.
- ☐ f is increasing for $x > 3$.
 f is decreasing for $x < 3$.

✔ Correct
First derivative: $x^3 - 3x^2 - 9x + 4$ and solve by equating to 0, we get $3x^2 - 6x - 9 = (x - 3)(x + 1)$ and $x = 3, x = -1$

$$f'(4) = 3x^2 - 6x - 9 = 3 * 4^2 - 6 * 4 - 9 = 15$$

$$f'(-2) = 3x^2 - 6x - 9 = 3 * (-2)^2 - 6 * (-2) - 9 = 15$$

f is increasing for $x < -1, x > 3$

f is decreasing for $-1 < x < 3$

3. Consider $f(x) = \frac{1}{x}$, where is the function increasing and decreasing?

1 / 1 point

- ☒ f is decreasing everywhere it is defined
- ☐ f is increasing everywhere it is defined.
- ☐ f is increasing for $x < 0$.
 f is decreasing for $x > 0$.

✔ Correct
 $f(x)$ is decreasing for $x < 0$ and $x > 0$

Think of $f(-2), f(-1)$ and $f(2), f(1)$ and at $x = 0, f(x)$ is undefined.

